
Campus Resilience: Modeling Human Evacuation via POMDP

in Extreme Weather Scenarios - A Case Study at the University of Utah

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Abstract

Emergency evacuation on university campuses presents a uniquely challenging decision-making problem: agents must navigate complex, hazard-prone terrain under severe perceptual limitations and time pressure. The University of Utah, situated at the foothills of the Wasatch Range near one of the most seismically active fault systems in the western United States, faces compounding risks from both winter blizzards and earthquake events. Despite this, current campus emergency plans rely on static routing tools that assume full environmental observability and passive human agents—assumptions that fail under real disaster conditions. This paper presents a case study modeling large-scale human evacuation at the University of Utah as a Partially Observable Markov Decision Process (POMDP). We make three contributions: (1) the first formal POMDP-based framework for outdoor campus evacuation under compound hazard scenarios, (2) a high-fidelity terrain-aware simulation environment integrating OSMnx street graphs with USGS slope and snow-depth data, and (3) a heterogeneous agent population model that surfaces bottlenecks and vulnerable subpopulations invisible to homogeneous baselines. Agent policies are trained using Deep Recurrent Q-Networks (DRQN), which maintain an implicit belief state via an LSTM memory module to overcome perceptual limitations imposed by extreme weather.

1 Introduction

The University of Utah campus sits at the foothills of the Wasatch Range, a setting that renders it uniquely vulnerable to compounding environmental hazards. Recurrent winter blizzards reduce visibility, increase slip risk on sloped pathways, and disrupt standard pedestrian routes. More critically, the campus lies in close proximity to the Wasatch Fault, one of the most seismically active fault systems in the western United States. The Utah Seismic Safety Commission estimates a 43% probability of a magnitude 6.75 or greater earthquake, and a 57% probability of a magnitude 6.0 or greater earthquake, occurring within the next 50 years [11, 12]—events that could simultaneously block evacuation routes, damage infrastructure, and generate mass-casualty scenarios.

Despite this well-documented risk profile, current campus emergency preparedness plans rely predominantly on static maps and broadcast notifications, tools that assume a fully observable environment and a passive human agent. Real evacuation, however, is a dynamic, partially observable process: individuals must make sequential, time-critical decisions based on incomplete sensory input, uncertain terrain conditions, and limited communication. The majority of existing evacuation models, including those based on social force models, cellular automata, and graph-search heuristics, operate under a full observability assumption that fails to reflect this perceptual reality. Prior work applying reinforcement learning to evacuation primarily considers indoor environments with well-defined exits and homogeneous agents; outdoor, multi-terrain, heterogeneous scenarios such as a university campus during a blizzard or earthquake remain underexplored.

This project addresses this gap by modeling campus evacuation as a Partially Observable Markov Decision Process (POMDP), a principled framework for sequential decision-making under uncertainty [5]. Agents maintain a belief state—a probability distribution over possible world configurations—which they update as new observations arrive, enabling principled trade-offs between speed, safety, and information-gathering.

This project makes three primary contributions:

1. **DRQN-based POMDP for Outdoor Campus Evacuation.** We present the first application of Deep Recurrent Q-Networks within a POMDP framework to model large-scale outdoor campus evacuation under compound hazard scenarios, including blizzards and seismic events, at the University of Utah. By maintaining an implicit belief state through LSTM memory, agents learn policies that are robust to severe perceptual limitations inherent in real-world extreme weather and disaster conditions.
2. **High-Fidelity Terrain-Aware Simulation Environment.** We construct a physics-grounded simulation environment of the University of Utah campus by integrating OSMnx street graph data [1] with USGS slope elevation profiles and empirical snow-depth models. The resulting environment captures slip risk and metabolic cost across real campus terrain, providing a more realistic testbed than prior indoor or topology-only evacuation simulations.
3. **Heterogeneous Agent Population Modeling.** We model a diverse agent population with varying physical capabilities and risk tolerances, including students, elderly staff, and mobility-impaired individuals, and systematically analyze how population heterogeneity affects collective evacuation outcomes. This analysis surfaces bottlenecks and high-risk subpopulations that homogeneous models would fail to identify.

2 Related Work

2.1 Evacuation Modeling

Pedestrian evacuation has been extensively studied through physics-inspired and graph-based approaches. Social force models [4] simulate agents as particles subject to repulsive and attractive forces, capturing crowd dynamics such as congestion and lane formation. Cellular automata models discretize the environment into grid cells and apply local transition rules to simulate flow. While computationally efficient, both paradigms assume agents have full knowledge of the environment and operate under static, pre-defined route assignments. These assumptions break down in outdoor disaster scenarios where terrain conditions evolve dynamically and visibility is severely limited. More recent work has explored graph-based route optimization [6] and agent-based simulation for building evacuation [2], but outdoor, multi-hazard, campus-scale scenarios remain largely unaddressed.

2.2 Reinforcement Learning for Disaster Response

Reinforcement learning has emerged as a promising paradigm for sequential decision-making in disaster response and navigation under uncertainty. Deep Q-Networks [8] demonstrated that RL agents can learn complex policies directly from raw observations, while Hausknecht and Stone [3] extended this to partially observable settings through Deep Recurrent Q-Networks (DRQN), which use an LSTM layer to integrate temporal observations into an implicit belief state. Sharma et al. [9] proposed the first dedicated RL environment for fire evacuation, training DQN agents on graph-based building models with realistic features such as fire spread and bottlenecks, demonstrating that RL

can significantly outperform shortest-path baselines. However, these efforts remain limited to fully observable, single-hazard, indoor settings, and do not address the compound partial observability imposed by blizzard conditions or post-seismic infrastructure failure in outdoor environments.

2.3 Heterogeneous Multi-Agent Simulation

Large-scale multi-agent simulation has advanced significantly with frameworks such as AgentSociety [10], which supports the simulation of thousands of heterogeneous agents with diverse behavioral profiles in urban environments. Research on heterogeneous evacuation populations has highlighted that mobility constraints, age, and risk perception significantly affect both individual and collective outcomes [7]. Studies of real evacuation events, including the 2011 Tohoku earthquake, consistently show that homogeneous agent assumptions lead to underestimates of evacuation time and misidentification of high-risk subpopulations. Our work builds on these findings by explicitly parameterizing agent heterogeneity and studying its effect on DRQN policy performance in a POMDP setting.

3 Methodology

Inspired by the AgentSociety framework [10], this project focuses on the large-scale interaction of heterogeneous agents across a realistically modeled outdoor environment.

3.1 GIS and Environmental Modeling

Using OSMnx [1], we generate a high-fidelity directed graph of the University of Utah campus street and pedestrian network. Each edge is annotated with slope data derived from USGS digital elevation models to simulate the metabolic cost and slip risk associated with Utah’s terrain under snow and ice conditions. Dynamic edge weights are updated during simulation to reflect real-time blockage events such as downed structures or snow drifts.

3.2 RL Architecture

We model the campus evacuation task as a POMDP defined by the 7-tuple $(S, A, T, R, \Omega, O, \gamma)$, where each component is grounded in the physical and operational conditions of the University of Utah campus.

- **States (S):** A state $s \in S$ encodes the full campus configuration at a given time step, including the occupancy and blockage status of each graph edge, per-edge slope friction coefficients (derived from USGS elevation data), current snow depth, structural integrity of buildings, and the position and physical condition of every agent. This state space is high-dimensional and not directly observable by any single agent.
- **Actions (A):** At each time step, an agent selects from a discrete action set: moving to an adjacent graph node, seeking shelter in a nearby building, or waiting in place. The action space is agent-specific, as mobility-impaired agents have access to a restricted subset of movement actions reflecting physical constraints.
- **Transitions (T):** $T(s' | s, a)$ defines the probability of transitioning to state s' after taking action a in state s . Transition probabilities are stochastic: a movement action may fail due to ice-induced slipping (modeled as a function of slope angle and snow depth) or route blockage from seismic damage. Environmental conditions such as snow accumulation and aftershocks evolve dynamically between time steps.
- **Observations (Ω):** Each agent receives a local observation $o \in \Omega$ rather than the full state. In blizzard conditions, visibility is limited to approximately 5 meters, restricting an agent’s awareness to immediately adjacent nodes. In seismic scenarios, agents additionally receive noisy structural integrity signals from nearby buildings. Observations do not include global path status or the positions of other agents.
- **Observation Function (O):** $O(o | s', a)$ defines the probability of receiving observation o after transitioning to state s' via action a . Observation noise is parameterized by weather severity: under whiteout conditions, even adjacent edge status may be perceived incorrectly, requiring agents to rely more heavily on their accumulated belief state rather than current sensory input.

- **Reward (R):** The reward function is multi-objective:

$$R = w_s \cdot \text{Survival} - w_t \cdot \text{Time} - w_e \cdot \text{Exposure}$$

Survival delivers a large positive reward upon reaching a designated safe zone. Time penalizes each elapsed time step to incentivize efficiency. Exposure penalizes time spent on high-risk edges (steep, icy, or structurally compromised). The weights w_s , w_t , and w_e are treated as hyperparameters and varied across experiments to analyze trade-offs between survival probability, evacuation speed, and hazard avoidance.

- **Discount Factor (γ):** A discount factor $\gamma \in [0, 1]$ controls the agent’s temporal planning horizon. A value close to 1 encourages long-horizon planning, appropriate for campus-scale evacuation where safe zones may require sustained navigation over many time steps.

Since the true state s is not directly accessible to agents, we train policies using Deep Recurrent Q-Networks (DRQN) [3]. The LSTM layer within DRQN maintains a hidden state across time steps, enabling each agent to accumulate a history of partial observations into an implicit belief state. This allows the agent to reason about unobserved portions of the campus (e.g., blocked paths not yet encountered) and make informed decisions under perceptual uncertainty. Agents are initialized with heterogeneous parameters including maximum movement speed, slip susceptibility, and risk aversion coefficient, reflecting the diverse population of students, staff, and mobility-impaired individuals present on campus.

4 Experiments

Agent policies are evaluated along three axes: (1) survival rate across a range of hazard intensities, (2) mean evacuation time, and (3) exposure time to dangerous conditions. We compare DRQN policies against two baselines: a shortest-path router (which assumes full observability) and a random-walk agent. We additionally vary the reward weights w_s , w_t , w_e to analyze the trade-off surface between survival, speed, and hazard avoidance, and run ablations across agent population compositions to measure the effect of heterogeneity on collective outcomes.

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